

Bubonic Plague (MONDO:0001112): Comprehensive Disease Characterization

Autonomous scientific discovery report — 5 iterations, 13 confirmed findings, 35 papers reviewed

Disease: Bubonic Plague · **MONDO:** MONDO:0001112 · **Category:** Infectious Disease **Causal agent:** *Yersinia pestis* (NCBI:txid632) **Evidence base:** Aggregated disease-level resources + primary literature (PubMed). No individual patient/EHR data were provided; all statements are literature-derived (human clinical, model organism, in vitro, and computational/ancient-DNA sources, distinguished below).

Summary

Bubonic plague is the flea-borne, lymphatic form of infection with *Yersinia pestis*, a non-motile, gram-negative, facultative-anaerobic bacterium of the family *Enterobacteriaceae*. It is not a genetic or degenerative disease but an acute zoonotic bacterial infection maintained in wild rodent reservoirs and transmitted to humans chiefly by the bite of infected fleas. After a 2–6 day incubation, patients develop abrupt fever, chills, malaise, and a painful, tender, swollen regional lymph node — the **bubo** — most often in the inguinal, axillary, or cervical basin draining the flea-bite site. Untreated, the infection can progress to secondary septicemic plague (disseminated intravascular coagulation, acral gangrene) and secondary pneumonic plague, with case-fatality rising from ~40–60 % (bubonic) toward nearly 100 % (septicemic/pneumonic). Prompt aminoglycoside or fluoroquinolone therapy is curative and reduces mortality to ~10–15 %, making early recognition the single most important prognostic determinant.

Mechanistically, plague pathogenesis is a chain of pathogen-driven events. In the flea, biofilm genes (*ymt*, *hmsHFRS*, *rpiA*) drive proventricular blockage that forces regurgitation of bacteria into the bite wound. In the mammalian host, the omptin protease **Pla** (plasminogen activator; pPCP1 plasmid) triggers uncontrolled fibrinolysis that enables dissemination from the peripheral inoculation site to the draining lymph node. There, the temperature-induced **Type III secretion system** (T3SS, pCD1/pYV plasmid) injects Yops and presents LcrV to disable phagocytes, permitting bacterial replication that forms the bubo and seeds the bloodstream

and lungs. *Y. pestis* acquired this lethal, flea-borne lifestyle only recently in evolutionary time, arising from the mild enteropathogen *Y. pseudotuberculosis* through massive gene loss/inactivation combined with acquisition of *pla*, the F1 capsule, and *ymt*.

Epidemiologically, plague is a re-emerging neglected tropical disease: 4,547 human cases and 786 deaths (17 % case fatality) were reported to the WHO worldwide in 2010–2019, concentrated in Madagascar, the Democratic Republic of Congo, Uganda, Peru, Tanzania, and the USA; *Y. pestis* was added to the WHO priority/pandemic pathogen list in 2024. Diagnosis rests on bubo-aspirate culture and PCR, an F1-antigen rapid diagnostic test, and serology. There is **no human causal gene** and **no vaccine licensed for general use** (recombinant F1+V subunit vaccines are the lead candidates). Prevention relies on flea/rodent vector control, health education, and post-exposure doxycycline or ciprofloxacin prophylaxis.

Section 1 — Disease Information

Overview. Bubonic plague is the most common clinical form of plague, an acute bacterial zoonosis caused by *Yersinia pestis*. "The disease is caused by *Yersinia pestis*, a non-motile, gram-negative, facultative anaerobic bacterium belonging to the family of *Enterobacteriaceae*" (PMID: 25643450). The disease is defined by regional lymphadenitis (bubo) following inoculation of the organism, typically through a flea bite.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0001112 (bubonic plague)
ICD-10	A20.0 (bubonic plague); A20 (plague)
ICD-11	1B93.0 (bubonic plague)
MeSH	D010930 (Plague) / "Plague, Bubonic"
SNOMED CT	Bubonic plague (disorder)
Pathogen (NCBI Taxonomy)	<i>Yersinia pestis</i> , txid632

OMIM and Orphanet are not applicable as primary sources because plague is an infectious, non-Mendelian disease; OMIM/Orphanet host-genetics entries do not define the condition.

Synonyms / alternative names. Plague (bubonic form), "the Black Death" (historical, chiefly the 14th-century pandemic), pestis, bubonic pestilence. Related clinical forms of the same infection: septicemic plague, pneumonic plague, pharyngeal and meningial plague.

Data provenance. Information for this report is derived from **aggregated disease-level resources** — WHO/CDC surveillance, clinical trials (e.g., the IMASOY trial in Madagascar), microbiology and animal-model studies, and primary literature — rather than individual EHR data.

Section 2 — Etiology

Primary cause (infectious). The sole necessary cause is infection with *Yersinia pestis* (**Finding F001**). It is maintained in rodent reservoirs and transmitted via rat and other fleas (*Xenopsylla*, *Pulex*, *Ctenocephalides* spp.); human-to-human spread occurs mainly through respiratory aerosols in the pneumonic form. Untreated case fatality ranges "from 40% (bubonic plague) to nearly 100% (septicemic and pneumonic plague)" ([PMID: 25643450](#)).

Environmental / occupational risk factors. Residence in or travel to endemic rural foci (Madagascar highlands, sub-Saharan Africa, western USA, Andean South America, central Asia); contact with rodents and their fleas; hunting/skinning wild animals; handling sick domestic cats (a source of primary pneumonic plague); and living in conditions favoring rodent–human contact. Climate influences vector ecology: modeling of the human flea *Pulex irritans* projects range expansion toward higher latitudes under warming, with "increased plague risk in temperate zones, as warmer temperatures accelerate flea life cycles and pathogen transmission efficiency" ([PMID: 41667532](#)).

Genetic risk / protective factors (host). There is no Mendelian susceptibility gene. Ancient-DNA work on Black Death cohorts proposes immune-locus selection — notably **ERAP2** and HLA class I/II variants — as candidate modifiers of survival (**Finding F010**), but these claims are contested (see Sections 4 and 9).

Gene–environment interactions. Not established for humans. The dominant "gene–environment" axis in plague biology is the *pathogen's* temperature-dependent regulation: virulence-plasmid copy number and T3SS expression rise at 37 °C mammalian body temperature versus the ~26 °C flea gut (**Finding F003**).

Section 3 — Phenotypes

Bubonic plague is a stereotyped acute febrile syndrome (**Finding F009**). Onset is **adult and all-age, acute**, 2–6 days after a flea bite.

Phenotype	Type	HPO suggestion	Frequency / notes
Fever (often abrupt, high)	Symptom/ sign	HP:0001945 (Fever)	Near-universal
Regional lymphadenopathy — the bubo (tender, swollen, painful node)	Clinical sign	HP:0002716 (Lymphadenopathy)	Defining feature; inguinal/femoral > axillary > cervical by bite site
Chills / rigors	Symptom	HP:0025143 (Chills)	Common
Headache	Symptom	HP:0002315 (Headache)	Common
Malaise / prostration	Symptom	HP:0033834 (Malaise)	Common
Hypotension / shock (septicemic progression)	Sign	HP:0002615 (Hypotension)	With dissemination
Disseminated intravascular coagulation	Lab/ clinical	HP:0005521 (DIC)	Septicemic form
Acral necrosis / gangrene ("black" digits)	Physical	HP:0100758 (Gangrene)	Late septicemic
Cough / dyspnea / hemoptysis (secondary pneumonic)	Sign	HP:0012735 (Cough); HP:0002105 (Hemoptysis)	If lungs seeded

Characteristics. Severity: moderate-to-severe, variable. Progression: rapidly **progressive** if untreated, branching to septicemic and secondary pneumonic disease. Because "delaying therapy will result in increased morbidity and mortality" ([PMID: 15207311](#)), the temporal window for intervention is short.

Quality-of-life impact. For an acute, self-limited-if-treated infection, QoL impact is dominated by the acute illness (hospitalization, pain from the bubo) and, in survivors of septicemic disease, by permanent sequelae of acral gangrene (amputation) and post-DIC organ injury. Formal EQ-5D/SF-36 data specific to plague were not identified.

Section 4 — Genetic / Molecular Information

Host causal genes: none. Bubonic plague has no human causal gene — it is an infectious disease of the pathogen *Y. pestis* (**Finding F010**). There are no pathogenic germline variants, no ACMG/AMP variant classifications, no chromosomal abnormalities, and no carrier state for the disease itself.

Candidate host modifier loci (contested). Ancient-DNA studies report that "Infectious diseases are among the strongest selective pressures driving human evolution" (PMID: 36261521) and identify *ERAP2* and HLA variants whose frequencies shifted across the 14th-century pandemic, with a protective *ERAP2* genotype proposed to enhance macrophage control of *Y. pestis*. Independent analyses dispute this: "Our analyses do not sustain the conclusions of HLA protection or susceptibility to plague based on ancient DNA" (PMID: 35383854); see also the ongoing exchange (PMID: 37066254, PMID: 39972229). These are host-genetics hypotheses, not disease-defining mutations.

Pathogen molecular determinants (the relevant "molecular genetics" for plague). Virulence is encoded on three plasmids plus the chromosome (**Findings F003, F005, F006, F007, F012**):

Plasmid	Key gene(s)	Product / role
pCD1 / pYV (~70 kb)	T3SS operons, <i>yopM</i> , <i>lcrV</i>	Type III secretion; injects Yops, presents LcrV; disables phagocytes
pPCP1 (~9.5 kb)	<i>pla</i>	Omptin protease (plasminogen activator); dissemination
pMT1 (~100 kb)	<i>caf1</i> (F1 capsule), <i>ymt</i>	Antiphagocytic capsule; murine toxin/phospholipase D for flea survival
Chromosome	<i>hmsHFRS</i> , <i>rpiA</i>	Biofilm exopolysaccharide; proventricular blockage

Epigenetic information. Not applicable to the host in a disease-defining sense.

Section 5 — Environmental Information

Infectious agent. *Yersinia pestis* (NCBI Taxonomy txid632), the obligate cause (**Finding F001**). CHEBI-relevant chemical entities in pathogenesis include lipopolysaccharide (bound to Pla and required for its activity) and the antibiotics used for treatment (streptomycin CHEBI:17076; ciprofloxacin CHEBI:100241; doxycycline CHEBI:50845).

Vectors and reservoirs. Fleas are the environmental transmission vehicle. Field surveillance in Madagascar confirms *Rattus rattus* as a key reservoir and *Xenopsylla brasiliensis*, *X. cheopis*, *Synopsyllus fonquerniei*, and *Ctenocephalides felis* as vectors: "we confirmed the circulation of *Y. pestis* ... one rat seropositive and one flea PCR positive ... *R. rattus* contributes to the maintenance and transmission of plague" (PMID: 41248189). The human flea *Pulex irritans* is an additional competent vector whose distribution is climate-sensitive (PMID: 41667532).

Environmental / lifestyle factors. Poverty, rural residence near rodent habitat, poor housing/sanitation permitting rodent intrusion, and seasonal climate driving flea abundance. Toxin/radiation/pollution exposures are not relevant.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating event → clinical manifestation)

1. An infected flea feeds; **proventricular biofilm blockage** (driven by *ymt*, *hmsHFRS*, *rpiA*) prevents normal blood passage → **leads to** regurgitation of *Y. pestis* into the dermal bite wound (**Finding F006**).
2. Deposited bacteria at ~37 °C **result in** rapid up-regulation of temperature-dependent virulence programs, including elevated pYV plasmid copy number and T3SS expression (**Finding F003**).
3. The omptin protease **Pla** activates plasminogen to plasmin and urokinase while inactivating α 2-antiplasmin/PAI-1 → **leads to** uncontrolled fibrinolysis and reduced fibrin(ogen) deposition, which **enables** bacterial migration from the peripheral site to the **draining lymph node** (**Finding F007**; supported by in vivo mouse evidence).
4. Within the node, the **T3SS injects Yops and presents LcrV** → **disables** phagocyte function (phagocytosis, oxidative burst, cytokine signaling), **allowing** unchecked bacterial replication (**Finding F003**).

5. Nodal replication plus the antiphagocytic F1 capsule **produce** the enlarging, inflamed, painful **bubo** — the defining clinical lesion (**Finding F009**).
6. **Branch A (septicemic)**: bacteria spill into blood → bacteremia, endotoxin-driven inflammation, **DIC**, hypotension, and acral **gangrene** (**Findings F009, F013**).
7. **Branch B (secondary pneumonic)**: hematogenous seeding of the lungs → pneumonia that is transmissible person-to-person by aerosol, closing the loop to new epidemics (**Findings F009, F013**).

Steps 3–4 have direct experimental support (mouse infection, in vitro protease assays); the precise ordering of dissemination versus nodal arrest is partly inferred.

Detail by category

- **Molecular pathways / biochemistry.** Pla-driven activation of the host **plasminogen–plasmin fibrinolytic cascade**: "Pla activates human plasminogen to the serine protease plasmin and activates the physiological plasminogen activator urokinase" ([PMID: 23898467](#)). These functions "enhance uncontrolled fibrinolysis ... and lowered fibrin(ogen) deposition has indeed been observed in mice infected with Pla-positive *Y. pestis*."
- **Protein dysfunction (pathogen effectors).** Pla is an outer-membrane β -barrel omptin whose activity depends on bound LPS; LcrV is "a key virulence factor of *Yersinia pestis* type III secretion system" ([PMID: 41724296](#)). Pathogenic *Yersiniae* "require the type III secretion system (T3SS) virulence factor to subvert host defense mechanisms and colonize host tissues" ([PMID: 40424556](#)).
- **Immune subversion / inflammation.** Yop effectors suppress phagocytosis and innate signaling; systemic dissemination triggers cytokine storm and coagulopathy.
- **Tissue-damage mechanisms.** Lymphadenitis with hemorrhagic necrosis (bubo); ischemic necrosis/gangrene from DIC microthrombosis; hemorrhagic pneumonia.
- **Temperature regulation.** "The number of pYV molecules relative to the number of chromosomes per cell ... increases with temperature" ([PMID: 40424556](#)), coupling the flea→mammal transition to virulence induction.

Suggested ontology terms. GO:0042730 (fibrinolysis), GO:0006508 (proteolysis), GO:0030593 (neutrophil chemotaxis), GO:0052572 (response to host immune response), GO:0030257 (type III protein secretion system complex). Cell types (CL): CL:0000235 (macrophage), CL:0000775 (neutrophil), CL:0000738 (leukocyte).

Section 7 — Anatomical Structures Affected

- **Primary organ / tissue:** the regional **lymph node** draining the inoculation site (UBERON:0000029, lymph node) — the bubo. Most commonly inguinal/femoral (UBERON:0035328), then axillary (UBERON:0002105), then cervical.
 - **Secondary organ involvement:** blood/vasculature (septicemia; DIC), skin and acral extremities (gangrene; UBERON:0002097 skin), **lung** (UBERON:0002048; secondary pneumonic plague), spleen and liver (bacterial seeding), and rarely meninges (plague meningitis; UBERON:0002360).
 - **Body systems:** lymphatic/immune, cardiovascular (coagulopathy, shock), integumentary, respiratory.
 - **Cell populations targeted:** phagocytes — macrophages (CL:0000235) and neutrophils (CL:0000775) — which the T3SS disables; endothelium is injured indirectly via coagulopathy.
 - **Subcellular:** Pla resides in the bacterial outer membrane; the T3SS forms an injectisome bridging pathogen and host-cell cytoplasm (GO:0030257).
 - **Lateralization:** the bubo is typically **unilateral**, on the side of the draining basin nearest the bite.
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Section 8 — Temporal Development

- **Onset:** acute; **incubation 2–6 days** after flea bite (**Finding F009**). All ages affected.
 - **Progression / stages:** (i) early localized febrile lymphadenitis (bubo); (ii) systemic dissemination → secondary septicemic plague; (iii) secondary pneumonic plague; (iv) end-stage multiorgan failure/shock. Progression rate is **rapid** (hours to days).
 - **Course:** monophasic, self-limited **only** with prompt treatment; otherwise progressive and frequently fatal. Not relapsing-remitting or chronic.
 - **Critical period:** the therapeutic window is narrow — survival hinges on antibiotics started early, ideally within the first 24 hours of symptoms.
 - **Remission:** treatment-induced recovery is the rule when therapy is timely; spontaneous resolution of untreated bubonic disease is uncommon (~40–60 % untreated mortality).
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Section 9 — Inheritance and Population (Epidemiology)

Epidemiology (Findings F004, F009). Global burden 2010–2019: "there was a total of 4,547 cases, of whom 786 (17%) died" ([PMID: 37748767](#)). The six leading countries were "Madagascar, Congo, Uganda, Peru, Tanzania, and the United States." Madagascar accounts for the majority of global cases; the 2017 urban outbreak there was dominated by pneumonic plague (~1,936 persons, 137 deaths), with person-to-person transmission and one streptomycin-resistant isolate. *Y. pestis* is now on the WHO priority pathogen list: "the 2024 updated WHO list of priority pathogens also recognizes ... *Yersinia pestis*" ([PMID: 40022523](#)).

Inheritance: not applicable (infectious disease; no Mendelian inheritance, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier frequency for the disease). Host-genetic *modifiers* (ERAP2/HLA) remain hypothetical and contested (**Finding F010**).

Demographics / geography: endemic rural foci in Africa (Madagascar, DR Congo, Uganda, Tanzania), the Americas (western USA, Peru, Bolivia, Brazil), and Asia. Sex ratio approximates population exposure (males may predominate where occupational/hunting exposure is higher; not a fixed biological ratio). Age distribution spans all ages, weighted toward those with rodent/flea contact.

Section 10 — Diagnostics

Confirmatory testing (Finding F005). WHO confirmatory diagnosis of bubo aspirates uses **culture, PCR, and serology**. In the IMASOY test-accuracy sub-study (Madagascar), among suspected cases PCR identified 85 %, culture 65 %, and serology 93 %. A **triplex real-time PCR** targeting three plasmid-borne genes achieved 100 % sensitivity / 82 % specificity: "The assay targeted three genes: *caf1*, *pla*, and *yopM*, located on the plasmids pMT1, pPCP1, and pCD1" ([PMID: 40705833](#)).

Rapid diagnostic test. The F1-antigen lateral-flow rapid test (F1RDT) is the field workhorse: "The sensitivity and specificity of on-site F1RDT were 94% ... and 74% ... against RS1" ([PMID: 41389991](#)). Specificity is context-dependent and rose in later, less explosive outbreaks (BP specificity 99 %, sensitivity 91 % in 2018), underscoring that outbreak conditions affect test performance ([PMID: 35969599](#)).

Modality	Target	Sensitivity	Specificity	Use case
Culture (bubo aspirate, blood)	Live <i>Y. pestis</i>	~65 %	Gold standard	Confirmation, AST
PCR (triplex qPCR)	<i>caf1, pla, yopM</i>	~85–100 %	82 %	Confirmation
F1RDT (lateral flow)	F1 capsular antigen	94 %	74 % (up to 99 % later)	Field/point-of-care
Serology	Anti-F1 antibody	~93 %	High	Retrospective/convalescent

Sample & clinical criteria. Bubo aspirate, blood, and (for pneumonic) sputum. Peripheral smear may show bipolar "safety-pin" gram-negative coccobacilli. **Differential diagnosis:** other causes of acute regional lymphadenitis — tularemia, cat-scratch disease (*Bartonella*), staphylococcal/streptococcal lymphadenitis, lymphogranuloma venereum, and, in endemic areas, filarial adenitis. Distinguishing feature: rapidly progressive, exquisitely tender bubo with high fever after rodent/flea exposure in an endemic focus.

Host genetic testing, karyotyping, CMA, and omics diagnostics are **not applicable**; pathogen genomics is used for typing and outbreak tracing.

Section 11 — Outcome / Prognosis

Mortality (Findings F001, F009). Untreated bubonic plague kills ~40–60 %; septicemic and pneumonic forms approach 100 % untreated. With **prompt** antibiotics, bubonic case-fatality falls to ~10–15 %. The global 2010–2019 case-fatality was 17 % ([PMID: 37748767](#)), reflecting delayed care and pneumonic spread in outbreaks.

Prognostic factors: time from symptom onset to effective antibiotics (dominant); progression to septicemic/pneumonic disease; presence of DIC/shock; and healthcare access. "Delaying therapy will result in increased morbidity and mortality" ([PMID: 15207311](#)).

Recovery / sequelae: survivors treated early generally recover fully. Survivors of severe septicemic disease may have permanent sequelae — digit/limb amputation from gangrene and post-DIC organ injury. Plague is not chronic; there is no long-term carriage.

Section 12 — Treatment

First-line pharmacotherapy (Finding F002). "Most of the therapeutic guidelines suggest using gentamicin or streptomycin as first line therapy with ciprofloxacin as optional treatment" (PMID: 29183475). Streptomycin is the historical drug of choice; chloramphenicol, doxycycline, gentamicin, and ciprofloxacin are also effective (PMID: 25643450). Chloramphenicol is preferred for plague meningitis (CNS penetration).

Intracellular efficacy nuance. *Y. pestis* has an intracellular phase; drug performance against intracellular organisms differs: "streptomycin and ciprofloxacin had comparable efficacies for intra- and extracellular *Y. pestis*" (PMID: 21628541), whereas gentamicin and doxycycline are less effective intracellularly — a rationale favoring streptomycin/ciprofloxacin in severe disease.

Drug (class)	Role	NCIT suggestion
Streptomycin (aminoglycoside)	First-line; historical DOC	NCIT:C839
Gentamicin (aminoglycoside)	First-line	NCIT:C516
Ciprofloxacin (fluoroquinolone)	First-line/alt; PEP	NCIT:C387
Levofloxacin (fluoroquinolone)	FDA-approved for plague	NCIT:C1667
Doxycycline (tetracycline)	Treatment/PEP	NCIT:C480
Chloramphenicol	Meningitis, alt	NCIT:C328

Experimental / advanced therapeutics. Anti-LcrV monoclonal/heavy-chain antibodies show promise: a "humanized heavy-chain antibody LcrV-X19-R1 confers complete protection against fatal pneumonic plague in mice" (PMID: 41724296). No gene, cell, or RNA therapies apply (bacterial infection). **Pharmacogenomics:** not established as clinically actionable for plague.

Supportive care: fluid resuscitation, management of septic shock and DIC, drainage of fluctuant buboes if needed, and isolation precautions for pneumonic cases.

Section 13 — Prevention

Primary prevention (Finding F008). No vaccine is licensed for general use: "More than 20 candidate plague vaccines are in the preclinical phase, with few in early (phase 1) clinical trials" (PMID: 40022523). Lead candidates are **recombinant F1+V (rF1+rV) subunit vaccines**: macaques "immunised ... with an rF1+rV vaccine ... were fully protected against pneumonic plague following inhalational exposure to ... *Yersinia pestis* (strain CO92)," and protection required "immunity to both vaccine antigens" (PMID: 21570437). Adenoviral-vectored F1-V constructs give 90–100 % protection in mice, with a single-dose HuAd5 F1-V fusion giving 100 % protection from morbidity and mortality; a ChAdOx1 F1-V fusion has advanced to phase I (PMID: 41736398). Live attenuated *Y. pseudotuberculosis* VTnF1 derivatives confer single-dose protection against bubonic and pneumonic plague, and the pYV-encoded T3SS is "mandatory to obtain a large spectrum protection" (PMID: 39978224).

Public-health / vector control. Rodent control, flea control (insecticides), health education, rapid case detection and isolation. Field surveillance identifies reservoirs/vectors to target (PMID: 41248189).

Chemoprophylaxis (secondary/post-exposure). Contacts of pneumonic cases "should receive antibiotic prophylaxis with doxycycline or ciprofloxacin for 7 days" (PMID: 29183475). Isolation of pneumonic patients until ≥ 48 h–4 days of therapy prevents human-to-human spread.

Section 14 — Other Species / Natural Disease

Taxonomy of susceptible hosts. Plague is a **zoonosis** with a broad mammalian host range. Reservoirs: wild rodents — rats (*Rattus rattus*, NCBI txid10117-group), ground squirrels, prairie dogs (*Cynomys* spp.), marmots, gerbils. Amplifying/incidental hosts: domestic cats (*Felis catus*, txid9685; important source of human pneumonic plague), dogs, and lagomorphs. Field data confirm *R. rattus* as the key Madagascar reservoir (PMID: 41248189).

Vectors (arthropods). Fleas: *Xenopsylla cheopis*, *X. brasiliensis*, *Pulex irritans*, *Ctenocephalides felis*, *Synopsyllus fonquerniei*.

Comparative pathology. "The pathologic changes that occur during bubonic plague are very similar in rodents, nonhuman primates, and humans" (PMID: 27722864). Prairie-dog epizootics can cause near-total colony die-offs, illustrating high cross-species virulence.

Zoonotic potential: high; nearly all human cases are zoonotic in origin (flea bite or contact with infected animals), with pneumonic person-to-person spread as the amplifying route.

Section 15 — Model Organisms

Model systems (Finding F011). "Various types of animal models of plague have been developed, including mice, rats, guinea pigs, and nonhuman primates ... rodent and nonhuman primate models of pneumonic plague closely resemble the human disease and ... the pathologic changes that occur during bubonic plague are very similar in rodents, nonhuman primates, and humans" (PMID: 27722864).

Model	Type	Use / recapitulation
Mouse (<i>Mus musculus</i>)	Mammalian	Standard for virulence, vaccine efficacy, intradermal/aerosol challenge
Rat (<i>Rattus</i> spp.)	Mammalian	Bubonic model; reservoir biology
Guinea pig	Mammalian	Vaccine efficacy under Animal Rule
Cynomolgus macaque / African green monkey	NHP	Pneumonic plague closely mimics human; pivotal for licensure
Flea (<i>Xenopsylla</i> , <i>Oropsylla</i>)	Invertebrate vector	Transmission/biofilm biology (<i>ymt</i> , <i>hmsHFRS</i> , <i>rpiA</i>)

Regulatory relevance (Animal Rule). Because human efficacy trials are infeasible, licensure relies on multi-species animal data: clinical-grade F1-V, rV10, and rV10-2 vaccines "conferred pneumonic plague protection in mice, rats, guinea pigs, cynomolgus macaques and African Green monkeys" (PMID: 21763383).

Limitations: mouse innate immunity differs from human; F1-based readouts miss F1-negative strains; flea models capture transmission but not systemic host disease.

Mechanistic Model / Interpretation

The integrated model (Finding F013) unifies all findings into a single causal narrative:

EVOLUTION (recent):

Y. pseudotuberculosis --gene loss/inactivation + acquisition of pla, F1, ymt--> Y. pestis

FLEA STAGE (~26 C):

ymt + hmsHFRS + rpiA ---> proventricular biofilm blockage ---> regurgitation into bite wound

MAMMALIAN STAGE (37 C):

Temperature shift ---> up pYV copy number + T3SS expression

|

|

|

v

T3SS injects Yops + LcrV ---> phagocytes disabled ---> unchecked replication in LYMPH NODES

|

v

F1 capsule (antiphagocytic) ---> BUBO

|

+

Branch A: bacteremia

--> SEPTICEMIC (DIC, gangrene)

Branch B: lung seeding

--> secondary PNEUMONIA

(aerosol person-to-person)

Upstream vs downstream. Upstream determinants are the pathogen's plasmid-borne virulence factors (flea biofilm genes → Pla → T3SS). Downstream manifestations are the bubo, septicemia/DIC, and pneumonia. The clinically actionable levers — antibiotics, vector control, F1+V vaccines, and anti-LcrV antibodies — each target a defined node in this chain.

Evidence Base

PMID	Title (abbrev.)	Supports / role
25643450	<i>Yersinia pestis</i> and plague – an update	Agent, untreated CFR by form, drug options
29183475	Bichat guidelines... plague	First-line therapy; 7-day doxy/cipro PEP; isolation
21628541	<i>In vitro</i> efficacy... intracellular <i>Y. pestis</i>	Streptomycin/ciprofloxacin intracellular efficacy
40424556	Polyadenylase PAPI... virulence plasmid	T3SS essential; temperature-dependent pYV copy number
41724296	Anti-LcrV heavy-chain antibody	LcrV as T3SS virulence factor / therapeutic target
39978224	T3SS in live <i>Y. pseudotuberculosis</i> vaccine	F1 + T3SS required for broad protection
37748767	Plague... Second Decade	4,547 cases / 786 deaths / 17% CFR; geography
40022523	WHO priority pathogen editorial	WHO listing; >20 preclinical vaccines, none licensed
41389991	IMASOY diagnostics sub-study	F1RDT 94%/74%; culture/PCR/serology yields
40705833	Triplex qPCR for plague	caf1/pla/yopM targets; 100%/82%
32294143	Refined flea transmission model	ymt/hmsHFRS/rpiA proventricular blockage
23898467	Fibrinolytic activities of <i>Y. pestis</i>	Pla → plasmin/fibrinolysis → dissemination
21570437	rF1+V protects macaques	Subunit vaccine efficacy; both antigens needed
41736398	Adenoviral-vectored vaccine	90–100% aerosol protection; ChAdOx1 F1-V to phase I
15207311	Plague	Three clinical forms; treatment-timing prognosis
36261521	Immune genes and the Black Death	ERAP2/HLA selection hypothesis
35383854	Challenging aDNA HLA claims	Contests host-genetic selection claims
27722864		

PMID	Title (abbrev.)	Supports / role
	<i>Pathology and Pathogenesis of Y. pestis</i>	Model recapitulation; genome-reduction evolution
21763383	<i>rV10/F1-V in 5 species</i>	Multi-species Animal-Rule efficacy
41248189	<i>Plague in Makira, Madagascar</i>	Reservoir/vector confirmation
41667532	<i>Climate change and Pulex irritans</i>	Vector range expansion / plague risk
35969599	<i>Optimizing imperfect diagnostics, 2017</i>	Diagnostic test performance in outbreaks

Evidence source types: human clinical/epidemiological (37748767, 41389991, 40705833, 15207311, 25643450, 29183475, 35969599); animal-model (21570437, 41736398, 21763383, 27722864, 41724296, 39978224); in vitro/biochemical (23898467, 21628541, 40424556); field ecology (41248189, 41667532); ancient-DNA/computational (36261521, 35383854).

Limitations and Knowledge Gaps

- 1. Host-genetics controversy unresolved.** The ERAP2/HLA selection signal from Black Death aDNA (PMID: [36261521](#)) is directly disputed (PMID: [35383854](#), [39972229](#)); no validated human susceptibility/protective variant exists.
- 2. Vaccine gap.** No vaccine is licensed for general use; efficacy rests on animal data under the FDA Animal Rule, and F1-based approaches may not protect against F1-negative strains.
- 3. Antimicrobial resistance surveillance is thin.** Streptomycin-resistant isolates have appeared (2017 Madagascar); systematic AST/resistance-gene surveillance is limited.
- 4. Diagnostic specificity is outbreak-dependent.** F1RDT specificity varied from 74 % (explosive 2017 outbreak) to 99 % (2018), complicating case counts and burden estimates (PMID: [35969599](#)).
- 5. QoL and long-term sequelae** (post-gangrene, post-DIC) are essentially uncharacterized with standardized instruments.
- 6. First-line-therapy sourcing.** The original guideline PMID (15677847) content was corroborated via the Bichat guideline (PMID: [29183475](#)), which carries the identical recommendations.

7. **Human primary data are aggregated**, not individual-level (no EHR/cohort granularity in this report).

Proposed Follow-up Experiments / Actions

1. **Adjudicate host-genetic modifiers** with adequately powered, replicated cohorts and functional assays of the proposed ERAP2 protective genotype in human macrophage–*Y. pestis* infection models.
2. **Advance F1-V vaccines through phase II/III bridging** (e.g., ChAdOx1 F1-V) under the Animal Rule, and test constructs protective against F1-negative strains (add LcrV/T3SS antigens).
3. **Systematic AMR surveillance** across endemic foci, with whole-genome sequencing to track resistance determinants and streptomycin-resistant clones.
4. **Improve point-of-care diagnostics** — multiplex antigen/PCR devices robust to outbreak conditions, validated prospectively against culture/PCR reference standards.
5. **Clinical development of anti-LcrV immunotherapeutics** (e.g., LcrV-X19-R1) as adjuncts to antibiotics for severe/late-presenting disease.
6. **Climate-informed vector surveillance** targeting projected range-expansion zones for *Pulex irritans* and *Xenopsylla* spp. to pre-position response capacity.
7. **Standardized outcome/QoL follow-up** of survivors to quantify amputation and organ-injury sequelae.

Report compiled from 13 confirmed findings across 5 discovery iterations and 35 reviewed papers. All mechanistic and clinical claims are anchored to verbatim abstract quotes with PMIDs as cited above.
